Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/01/2008
PI Name:	Kostelecky, V. Alan Ph.D.		
Project Title:	Theoretical Studies of Lorentz and CPT Symmetry		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47405	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (5/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2914		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		
Task Description:	A general extension of the Standard Model of particle physics incorporating gravity, known as the Standard-Model Extension (SME), allows for the possibility of Lorentz and CPT violation while otherwise preserving conventional physics. Sensitive experiments searching for predicted effects include ones that may be advantageously performed in a microgravity environment. The project objectives are to develop and study theoretically possible violations of spacetime symmetry and to identify sensitive experimental tests for the associated effects. The project significance is that high-sensitivity tests of fundamental symmetries can shed light on physics at the Planck scale and could offer experimental support for an underlying theory such as strings.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The foundational theoretical study on Lorentz violation in the gravitational sector in the SME was published in 2004. During the reporting period, a major followup paper was published that takes advantage of the prospects opened up for tests in a variety of terrestrial and solar-system experiments and observations. This lengthy work first investigates the pure-gravity sector of the minimal SME in the limit of Riemann spacetime. The modifications to the Einstein field equations are obtained in the limit of small metric fluctuations about the Minkowski vacuum, while allowing for the dynamics of the 20 independent coefficients for Lorentz violation. The post-newtonian metric is extracted from the linearized effective equations, and the corresponding behavior of a perfect fluid is studied and applied to the gravitating many-body system. Illustrative examples of the methodology are provided using models of vectors with spontaneously broken Lorentz symmetry, known as bumblebee models. Following these theoretical preliminaries, the work investigates the implications for a variety of existing and proposed gravitational experiments, including both and satellite laser ranging, laboratory experiments with gravimeters and torsion pendula, measurements of the spin precession of orbiting gyroscopes including Gravity Probe B, timing studies of signals from binary pulsars, and the classic tests involving the perihelion precession and the time delay of light, including from the Cassini mission. Numerous effects of local Lorentz violation can be studied in existing or near-future experiments at interesting sensitivities. Several groups are presently undertaking experimental or observational searches for Lorentz violation based on the results presented in this work. The general theoretical framework also applies to the more general case of Riemann-Cartan spacetime, in which the torsion tensor can be nonvanishing. Effects due to torsion in Riemann-Cartan spacetime are expected to generate a variety of potentially observable signals. During the course of the grant, spectropolarimetry of cosmological sources was used to obtain stringent bounds on Lorentz violation comparable to the best current limits in the fermion sector. Ten of the 19 independent coefficients for Lorentz violation were bounded to less than 3x10^{-32}. In the current reporting period, another paper was published in Physical Review Letters, the leading letters journal in the field. A prior Physical Review Letter demonstrated that recent measurements of linear polarization in gamma rays from gamma-ray bursts can be used to constrain certain types of Lorentz violations in the photon sector to less than parts in 10^{-37}. The new work studies possible Lorentz-violating effects arising in the cosmic microwave background radiation. It develops a systematic classification of Lorentz-violating operators and uses spectropolarimetry to search for the associated effects.
Bibliography Type:	Description: (Last Updated: 03/05/2008)
Articles in Peer-reviewed Journals	Bailey Q, Kostelecky VA. "Signals for Lorentz violation in post-Newtonian gravity." Physical Review D 2006 Aug;74(4):045001. http://dx.doi.org/10.1103/PhysRevD.74.045001 , Aug-2006
Articles in Peer-reviewed Journals	Kostelecký VA, Mewes M. "Lorentz-violating electrodynamics and the cosmic microwave background." Physical Review Letters 2007 Jul 6;99(1):011601. PMID: 17678146 , Jul-2007